

Tier 1: Clients & Presentation Edge Gateway

Clinical Staff (React 19 UI)

Vite + TypeScript + Tailwind

- Doctors / Nurses (Role-Based JWT)
- Patient Census & Encounters Dashboard

Compliance & Audit Console

Admin & DPO Portal

- Statutory Retention & Override Auth
- Interactive Merkle Proof Verifier

Nginx 1.27 Edge Gateway | TLS 1.3 Termination | Port :443 / :80 | CSP Headers

REST API Reverse Proxy (TLS 1.3)

Tier 2: Application & Cryptographic Core (Spring Boot 3.3.4 / Java 21)

Security & RBAC Engine

Spring Security 6 Stateless

- RSA/ECDSA JWT Filter & Auth Guard
- Multi-Tenant Context Isolation

Envelope Encryption Core

AES-256-GCM + AAD Binding

- Ephemeral DEK Generation (256-bit)
- HMAC-SHA256 Blind Index Search

Verifiable Erasure & Hybrid Post-Quantum Signer

GDPR Art. 17 / NIST FIPS 204

- Rolling Retention Horizon $V_{active}(p)$ Evaluation & Statutory Override Enforcement
- Dual Proof Signatures: Vault RSA-2048 + Post-Quantum ML-DSA-65 & Merkle DAG Leaves

Coordinated 5-Stage Cryptographic Erasure Pipeline:

1. Auth
JWT Check

2. KMS Shred
DELETE KEK

3. DB Nullify
Null Blob

4. Redis Evict
Purge L2

5. Merkle Mint
Dual Sign

Transit KMS API (:8200)

Encrypted SQL / Kafka Events

Tier 3: Zero-Plaintext Persistence, KMS & Event Infrastructure

3-Node Vault Raft KMS

Port :8200 / Transit Engine

- Two-Tier Hierarchy (KEK_{pat} / KEK_{vis})
- Linearizable Raft Storage & Proof Signer

PostgreSQL 15 Datastore

Zero-Plaintext Relational Store

- AES-256-GCM Ciphertext Blobs
- Blind Index B-Trees & Merkle DAG Nodes

Redis 7 L2 Cache

In-Memory Acceleration

- 15-Minute TTL (TTL = 900s)
- Shredded-Entity Bypass Filter

Kafka 3.7 KRaft & Telemetry

Event Broker & Prometheus TSDB

- patient-record-events (1-Yr Retention)
- Prometheus & Grafana Mission Control